

Title

Feature Engineering, Model Optimization & Performance Comparison

Objective

To compare different machine learning regression models for house price prediction.

Dataset

California Housing Dataset.

Models Used

- Linear Regression
- Ridge Regression
- Decision Tree Regressor

Feature Engineering

- Data preprocessing
- Feature scaling using StandardScaler
- Train-test split

Evaluation Metrics

- RMSE (Root Mean Squared Error)
- R^2 Score

Results

Paste the comparison table screenshot.

Visualization

Paste the graph screenshot.

Conclusion

Mention which model performed best based on the highest R^2 score and lowest RMSE.